

Sl No	Description of Work	Units	Qty	Remark
j	Supply and installation of Pipe Insulation shall be of the pre-insulated type of polyurethane foam insulation with density of 48 kg/m3, Thermal conductivity of Polyurethane of 0.022 Kcal/mhr°C, Service temprature +5 deg (ambient temperature 35°C) with Compressive Strength of 2.4 kgf/cm2 min, Water vapour Permeability 3.7 perm. cm. and fire rating encased in a non-combustible casing having a fire resistance of 2 hours, all the chilled water pipe to be painted as per details shown in technical specifications and drawings.			Comply Density - 36 +10% Kg/m3 Compressive Strength - 115 KN/m2 Water absorption 1.6% by volume
	Elbow joints, tee joints and valve boxes etc are insulated by using the same materials and shall included in pipe insulation measurement			Noted
	The procedures for on-site insulation which is recommended by the manufacturer shall be strictly adhered.			
	Insulation of thickness shall follow as below:			
	40 mm for pipe up to 40 mm dia			
	50 mm for pipe up to 65 mm dia			
	55 mm for pipe up to 150 mm dia			
	66 mm for pipe up to 300 mm dia			
	76 mm for pipe above 350 mm dia			
	Casing shall be spirally formed locked seam galvanised steel having the spiral folded joints at 135 mm centres and the following thickness:			
	Casing size up to 210 mm diameter -0.5 mm thick (26 gauge)			
	Casing size 225 mm diameter and above - 0.6 mm thick (24 gauge)			
a	600 mm dia	Rmtr	144.00	
b	250 mm dia	Rmtr	150.00	
c	65 mm dia	Rmtr	9.00	
4	Supply and installation of Chilled Water Pipe Insulation shall be of the pre-insulated type of polyurethane foam insulation with density of 48 kg/m3, Thermal conductivity of Polyurethane of 0.022 Kcal/mhr°C, Service temprature +5 deg (ambient temperature 35°C) with Compressive Strength of 2.4 kgf/cm2 min, Water vapour Permeability 3.7 perm. cm. and fire rating encased in a non-combustible casing having a fire resistance of 2 hours, all the chilled water pipe to be painted as per details shown in technical specifications and drawings.			Comply Density - 36 +10% Kg/m3 Compressive Strength - 115 KN/m2 Water absorption 1.6% by volume
	Elbow joints, tee joints and valve boxes etc are insulated by using the same materials and shall included in pipe insulation measurement			Noted
	The procedures for on-site insulation which is recommended by the manufacturer shall be strictly adhered.			
	Insulation of thickness shall follow as below:			
	40 mm for pipe up to 40 mm dia			
	50 mm for pipe up to 65 mm dia			
	55 mm for pipe up to 150 mm dia			
	66 mm for pipe up to 300 mm dia			
	76 mm for pipe above 350 mm dia			
	Casing shall be spirally formed locked seam galvanised steel having the spiral folded joints at 135 mm centres and the following thickness:			
	Casing size up to 210 mm diameter -0.5 mm thick (26 gauge)			
	Casing size 225 mm diameter and above - 0.6 mm thick (24 gauge)			
a	25 mm dia	Rmtr	390.00	
b	32 mm dia	Rmtr	555.00	
c	40 mm dia	Rmtr	48.00	
d	50 mm dia	Rmtr	188.00	
e	65 mm dia	Rmtr	232.00	
f	80 mm dia	Rmtr	314.00	
g	100 mm dia	Rmtr		
h	125 mm dia	Rmtr		
i	150 mm dia	Rmtr		
j	200 mm dia	Rmtr	284.00	
k	250 mm dia	Rmtr		
l	300 mm dia	Rmtr	94.00	
m	350 mm dia	Rmtr	223.00	
n	400 mm dia	Rmtr	234.00	
o	450 mm dia	Rmtr	168.00	
p	500 mm dia	Rmtr	1,257.00	
q	600 mm dia	Rmtr	557.00	
s	750mm dia	Rmtr	-	
t	900 mm dia	Rmtr	573.00	
u	1100 mm dia	Rmtr		
25	Insulation for Chilled Pipes and Ducts.			
a	Duct Internal (Acoustic) Insulation as per technical specifications and drawings			
	Supply and installation of internal insulation shall be open cell elastomeric type. Class 1 nonflammable acoustic insulation of 15 mm (minimum) thick and 140- 180 kg/cum density. The material shall be self-extinguishing to ASTM635 and has a thermal conductivity of not more than 0.0465 w/mK at 25°C mean temperature. In addition, a flame retardant glass cloth layer shall be provided for all internal insulation used complete a sper technical specifications and drawings.	Sqmtr		Comply with deviation K value 0.0472 w/mk at 20 C
b	Duct Insulation as per technical specifications and drawings			
	Supply and Installation duct insulation materials of Class – O as per BS 476 with closed closed cell elastomeric insulation with foil, having a K factor of not more than 0.0332 W/degk at 24°C mean temperature. The insulation thickness shall be 15 mm for supply and 13 mm for return ducts and shall be of 40 - 60 kg/m3 nominal density complete as per technical specifications and drawings.			Comply with deviation k value 0.032 w/mk at 30 C
b.1	Insulation thickness with 15 mm for supply ducts	Sqmtr	5,702.00	
b.2	Insulation thickness with 13 mm for return ducts	Sqmtr	489.00	
b.c	Insulation thickness with 9 mm for return ducts located below the roof and pass through non-air conditioned areas	Sqmtr		
26	Weather exposed duct works			
	Supply and Installation duct insulation materials of Class – O as per BS 476, closed closed cell elastomeric insulation with foil, having a K factor of not more than 0.0332 W/degk at 24°C mean temperature. The insulation thickness shall be two layer of 7mm thick fiber glass cloth plus two layers of shield coating and shall be of 40 - 60 kg/m3 nominal density as per the specifications and detailed drawing.	Sqmtr		Comply with deviation k value 0.032 w/mk at 30 C

